

Statistical and Machine Learning Comparison for Predicting Patient Appointment No-Show Outcomes Using Healthcare Data

Author: Ayush Nagar

Position: Healthcare Data Analyst

Education: Master of Science (MSc) in Data Science

Academic Background: Bachelor of Science (BSc) in Statistics

Location: Jersey City, New Jersey, United States

Research Area: Healthcare Analytics, Statistical Modeling, Machine Learning, Predictive Modeling

Project Type: Independent Research Project

Year: 2026

Author Profile

Ayush Nagar is a Healthcare Data Analyst with a strong academic and professional background in statistics, data science, and healthcare analytics. He holds a Master of Science degree in Data Science and a Bachelor of Science degree in Statistics, providing him with a solid foundation in statistical inference, predictive modeling, and machine learning methodologies.

His research interests focus on the application of statistical modeling and artificial intelligence techniques to healthcare data in order to improve operational efficiency, optimize patient outcomes, and enhance predictive decision-making. His work integrates classical statistical methods such as hypothesis testing and regression modeling with modern machine learning approaches including ensemble learning and gradient boosting algorithms.

Through this research project, Ayush aims to contribute to the growing field of healthcare predictive analytics by evaluating the effectiveness of statistical and machine learning models in predicting patient appointment attendance outcomes. His work emphasizes both predictive performance and model interpretability, recognizing the importance of explainable models in healthcare decision-making environments.

His professional experience as a Healthcare Data Analyst provides practical insight into real-world healthcare operational challenges, including appointment scheduling inefficiencies, patient behavior prediction, and healthcare system optimization. This experience enables him to bridge the gap between theoretical statistical modeling and practical healthcare applications.

Research Objective

The primary objective of this study is to conduct a comprehensive comparison between traditional statistical modeling techniques and modern machine learning algorithms for predicting patient appointment no-show outcomes using healthcare operational data.

Specifically, this research aims to:

- Evaluate the predictive performance of logistic regression as a traditional statistical modeling approach
- Implement and evaluate machine learning models including Random Forest and XGBoost
- Compare model performance using multiple evaluation metrics including accuracy, confusion matrix analysis, and ROC curve analysis
- Identify the most important predictors influencing patient appointment attendance using feature importance analysis
- Assess the practical implications of predictive modeling for healthcare operational improvement

This study emphasizes both predictive accuracy and interpretability, recognizing the importance of transparent and explainable models in healthcare analytics.

Abstract

Healthcare appointment no-shows represent a persistent and costly challenge for healthcare systems worldwide. Missed appointments lead to inefficient utilization of healthcare resources, increased operational costs, longer patient wait times, and reduced overall system efficiency. Predicting patient appointment attendance has become an important application of healthcare analytics, enabling healthcare providers to implement preventive interventions such as automated reminders, targeted scheduling strategies, and patient engagement programs.

This study presents a comprehensive comparative analysis of traditional statistical modeling techniques and modern machine learning approaches for predicting patient appointment no-show outcomes using healthcare operational data. The dataset used in this research consists of 110,527 patient appointment records, including demographic, medical, and operational variables such as age, gender, scholarship status, hypertension status, diabetes status, and SMS reminder status. The outcome variable represents whether a patient attended their scheduled appointment or missed it.

The statistical analysis component of this study includes descriptive statistics, hypothesis testing, and logistic regression modeling. An independent samples t-test was conducted to evaluate whether age differed significantly between patients who attended appointments and those who did not. The results demonstrated a statistically significant difference in mean age between the two groups ($p < 0.001$), indicating that age is an important predictor of appointment attendance. Logistic regression analysis further confirmed that age is a statistically significant predictor, with a negative coefficient indicating that older patients are less likely to miss appointments compared to younger patients.

In addition to traditional statistical methods, machine learning models including Random Forest and Extreme Gradient Boosting (XGBoost) were implemented and evaluated. These models were trained and tested using a train-test split methodology to ensure proper model validation. The predictive performance of each model was evaluated using multiple performance metrics including classification accuracy, confusion matrix analysis, and Receiver Operating Characteristic (ROC) curve analysis.

The results showed that logistic regression achieved a predictive accuracy of 79.93%, while Random Forest achieved an accuracy of 79.71%, and XGBoost achieved an accuracy of 79.93%. These findings indicate that machine learning models did not provide a substantial improvement in predictive accuracy compared to the traditional statistical model. ROC curve analysis yielded an Area Under the Curve (AUC) score of 0.604 for the XGBoost model, indicating moderate predictive capability with the available predictor variables.

Feature importance analysis using the XGBoost model revealed that SMS reminder status was the most influential predictor of appointment attendance, followed by age. This finding highlights

the critical role of communication-based interventions in improving healthcare appointment attendance rates.

Overall, this study demonstrates that traditional statistical models remain highly effective and interpretable tools for healthcare predictive modeling, while machine learning models provide additional insights through feature importance analysis. The results of this research contribute to the understanding of predictive modeling applications in healthcare and provide practical insights for improving healthcare operational efficiency.

Introduction

Healthcare systems worldwide face significant operational challenges related to patient appointment management. One of the most persistent and costly issues is patient appointment no-shows, which occur when patients fail to attend scheduled medical appointments without prior cancellation. Appointment no-shows reduce the efficiency of healthcare delivery, lead to underutilization of clinical resources, increase administrative costs, and negatively affect access to care for other patients.

The impact of appointment no-shows extends beyond operational inefficiencies. Missed appointments disrupt clinical workflows, increase waiting times for other patients, and reduce the overall quality of healthcare service delivery. Healthcare providers must allocate staff, equipment, and facilities based on scheduled appointments, and no-shows result in wasted capacity and lost productivity. In large healthcare systems, appointment no-show rates can range from 10% to 30%, representing a significant economic burden.

Predictive analytics has emerged as a powerful approach for addressing healthcare operational challenges. By analyzing historical patient data, predictive models can identify patterns associated with patient behavior and estimate the probability that a patient will attend or miss an appointment. These predictions enable healthcare providers to implement targeted interventions, such as reminder notifications, rescheduling strategies, and patient engagement initiatives, to reduce no-show rates and improve healthcare system efficiency.

Statistical modeling has long been used as a foundational approach in healthcare analytics. Logistic regression, in particular, has been widely applied for predicting binary healthcare outcomes such as disease diagnosis, treatment response, and patient behavior. Logistic regression provides several advantages, including interpretability, statistical rigor, and the ability to quantify relationships between predictor variables and outcomes. The coefficients estimated by logistic regression models allow researchers and healthcare professionals to understand how individual variables influence the probability of a given outcome.

In recent years, machine learning techniques have gained increasing attention in healthcare predictive modeling. Machine learning algorithms such as Random Forest and Extreme Gradient Boosting (XGBoost) offer several advantages over traditional statistical models, including the ability to capture complex nonlinear relationships, model interaction effects, and improve predictive performance in certain scenarios. These models are capable of learning from large datasets and identifying patterns that may not be easily detected using traditional statistical approaches.

Random Forest is an ensemble machine learning method that constructs multiple decision trees and aggregates their predictions to produce a final prediction. This approach improves predictive accuracy and reduces overfitting compared to individual decision trees. Random

Forest models are particularly useful for handling datasets with complex variable interactions and nonlinear relationships.

Extreme Gradient Boosting (XGBoost) is an advanced gradient boosting algorithm that builds predictive models by sequentially combining weak learners to improve predictive performance. XGBoost has become widely used in predictive modeling due to its efficiency, scalability, and strong predictive performance across various applications, including healthcare analytics, financial modeling, and risk prediction.

Despite the growing popularity of machine learning methods, it is important to evaluate whether these advanced algorithms provide meaningful improvements over traditional statistical models in specific applications. In healthcare settings, model interpretability is often as important as predictive accuracy. Healthcare professionals must be able to understand and trust predictive models in order to use them effectively for clinical decision-making and operational planning.

This study aims to conduct a comprehensive comparison between traditional statistical modeling techniques and modern machine learning approaches for predicting patient appointment no-show outcomes. By implementing logistic regression, Random Forest, and XGBoost models on a real-world healthcare dataset, this research evaluates the predictive performance, interpretability, and practical implications of each modeling approach.

The dataset used in this study contains 110,527 patient appointment records, including demographic information, medical conditions, and operational variables. This large dataset provides a suitable foundation for both statistical and machine learning analysis. The study evaluates model performance using multiple evaluation metrics, including accuracy, confusion matrix analysis, and Receiver Operating Characteristic (ROC) curve analysis.

In addition to comparing predictive performance, this study also examines feature importance to identify the most influential factors affecting patient appointment attendance. Understanding which variables most strongly influence appointment attendance provides valuable insights for healthcare providers seeking to improve patient engagement and reduce no-show rates.

The findings of this research contribute to the growing field of healthcare predictive analytics by providing a detailed comparison of statistical and machine learning methods. The results offer practical insights for healthcare providers and demonstrate the strengths and limitations of different modeling approaches in predicting patient behavior.

This research highlights the importance of combining statistical rigor with modern machine learning techniques to develop effective predictive models for healthcare applications. By evaluating both statistical and machine learning methods, this study provides a balanced assessment of predictive modeling approaches and their practical value in healthcare operational optimization.

Dataset Description

The dataset used in this study consists of patient appointment records collected from a healthcare appointment scheduling system. The dataset contains a total of 110,527 observations, with each observation representing a scheduled appointment for a patient. The dataset includes demographic, medical, and operational variables that may influence whether a patient attends or misses their scheduled appointment.

The primary objective of this dataset is to provide information that can be used to predict patient appointment attendance behavior. Predictive modeling of appointment attendance is an important application of healthcare analytics, as it enables healthcare providers to identify patients at higher risk of missing appointments and implement preventive interventions.

Each observation in the dataset corresponds to a single appointment and contains multiple variables describing patient characteristics and appointment-related factors. The dataset includes a mixture of numerical and categorical variables, allowing for both statistical and machine learning analysis.

The outcome variable in this study is the appointment attendance indicator, labeled as "No-show." This variable is binary and represents whether a patient attended their scheduled appointment or did not attend. The variable was encoded as follows:

- No-show = 0 indicates that the patient attended the scheduled appointment
- No-show = 1 indicates that the patient did not attend the scheduled appointment

This binary outcome variable serves as the dependent variable in both statistical and machine learning models implemented in this study.

The dataset includes several predictor variables that may influence appointment attendance. These variables include demographic information such as patient age and gender, medical information such as the presence of chronic conditions, and operational information such as whether the patient received an SMS reminder prior to the appointment.

The Age variable represents the age of the patient at the time of the appointment and is measured in years. Age is a continuous numerical variable and is expected to influence appointment attendance behavior, as different age groups may exhibit different healthcare utilization patterns.

The Gender variable represents the patient's gender and is a categorical variable with two possible values: male and female. Gender differences may influence healthcare utilization patterns due to differences in health-seeking behavior, healthcare access, and social factors.

The dataset also includes a Scholarship variable, which indicates whether the patient is enrolled in a healthcare assistance program. This variable may reflect socioeconomic status, which can influence healthcare access and appointment attendance.

Two medical condition variables, Hypertension and Diabetes, are included in the dataset. These variables indicate whether the patient has been diagnosed with hypertension or diabetes, respectively. Patients with chronic medical conditions may have different healthcare utilization patterns compared to patients without such conditions.

The SMS_received variable indicates whether the patient received an SMS reminder prior to the scheduled appointment. SMS reminders are commonly used by healthcare providers to improve appointment attendance rates by reminding patients of upcoming appointments. This variable is particularly important for evaluating the effectiveness of communication-based interventions.

In addition to these predictor variables, the dataset includes several additional variables related to appointment scheduling, such as appointment date and scheduling date. While these variables provide additional contextual information, the primary focus of this study is on demographic, medical, and operational predictors.

Before conducting statistical and machine learning analysis, the dataset was examined to ensure data quality and completeness. Data preprocessing steps included verifying variable types, checking for missing values, and converting categorical variables into numerical format where necessary.

Descriptive statistics were computed to summarize the characteristics of the dataset. These statistics include measures such as mean, median, minimum, and maximum values for numerical variables, as well as frequency distributions for categorical variables.

The dataset contains a large sample size of over 110,000 observations, providing a strong foundation for predictive modeling. Large datasets improve the reliability of statistical estimates and allow machine learning models to learn meaningful patterns from the data.

Exploratory analysis of the outcome variable showed that the majority of patients attended their scheduled appointments, while a smaller proportion of patients missed their appointments. Specifically, approximately 79.8% of patients attended their appointments, while approximately 20.2% of patients did not attend their appointments.

This class distribution is important for predictive modeling, as it indicates that the dataset is moderately imbalanced, with more observations in the attended appointment category than in the missed appointment category. Class imbalance can influence model performance and must be considered when evaluating predictive accuracy.

The dataset provides a comprehensive set of variables that allow for meaningful analysis of patient appointment attendance behavior. By combining demographic, medical, and operational variables, the dataset enables both statistical and machine learning models to identify factors associated with appointment attendance.

The availability of communication-related variables such as SMS reminders provides an opportunity to evaluate the impact of healthcare communication strategies on patient behavior. This analysis has practical implications for healthcare providers seeking to improve appointment attendance rates and optimize healthcare system efficiency.

Overall, the dataset provides a suitable and robust foundation for conducting predictive modeling using both statistical and machine learning methods. The large sample size, diverse predictor variables, and clearly defined outcome variable make this dataset well-suited for evaluating predictive models and identifying factors influencing patient appointment attendance.

Exploratory Data Analysis

Exploratory Data Analysis (EDA) is a critical step in the data analysis process that involves examining and summarizing the main characteristics of a dataset before applying statistical or machine learning models. The primary purpose of exploratory data analysis is to understand the structure, distribution, and relationships within the dataset, identify patterns and anomalies, and ensure that the data is suitable for predictive modeling.

In this study, exploratory data analysis was performed to examine the distribution of predictor variables and the outcome variable, assess data quality, and identify potential relationships between variables and appointment attendance behavior.

The first step in exploratory data analysis involved examining the overall structure of the dataset. The dataset contains 110,527 observations and 14 variables, representing patient appointment records. Each observation corresponds to a scheduled appointment, and each variable provides information about patient characteristics or appointment-related factors.

A preview of the dataset was examined using the first few observations. This preview provides insight into the structure of the dataset, including variable names, data types, and example values. This step helps verify that the dataset has been loaded correctly and that variables are properly formatted.

[4]:

	PatientId	AppointmentID	Gender	ScheduledDay	AppointmentDay	Age	Neighbourhood	Scholarship	Hipertension	Diabetes	Alcoholism	Handcap
0	2.987250e+13	5642903	F	2016-04-29T18:38:08Z	2016-04-29T00:00:00Z	62	JARDIM DA PENHA	0	1	0	0	
1	5.589978e+14	5642503	M	2016-04-29T16:08:27Z	2016-04-29T00:00:00Z	56	JARDIM DA PENHA	0	0	0	0	
2	4.262962e+12	5642549	F	2016-04-29T16:19:04Z	2016-04-29T00:00:00Z	62	MATA DA PRAIA	0	0	0	0	
3	8.679512e+11	5642828	F	2016-04-29T17:29:31Z	2016-04-29T00:00:00Z	8	PONTAL DE CAMBURI	0	0	0	0	
4	8.841186e+12	5642494	F	2016-04-29T16:07:23Z	2016-04-29T00:00:00Z	56	JARDIM DA PENHA	0	1	1	0	

Figure 1. Dataset preview showing first five observations.

The dataset includes both numerical and categorical variables. Numerical variables such as Age represent continuous values, while categorical variables such as Gender and SMS_received represent binary or categorical information. Proper understanding of variable types is essential for selecting appropriate statistical and machine learning methods.

Descriptive statistics were computed to summarize the numerical variables in the dataset. These statistics include measures such as mean, standard deviation, minimum value, and maximum value. Descriptive statistics provide a summary of the central tendency and variability of numerical variables.

The Age variable is one of the most important predictors in the dataset. Analysis of the Age variable showed that patient ages range from young children to elderly individuals. The mean age of patients in the dataset is approximately 37 years, indicating that the dataset includes a wide range of adult patients.

To better understand the distribution of patient ages, a histogram was generated. The histogram shows the frequency distribution of age values across the dataset.

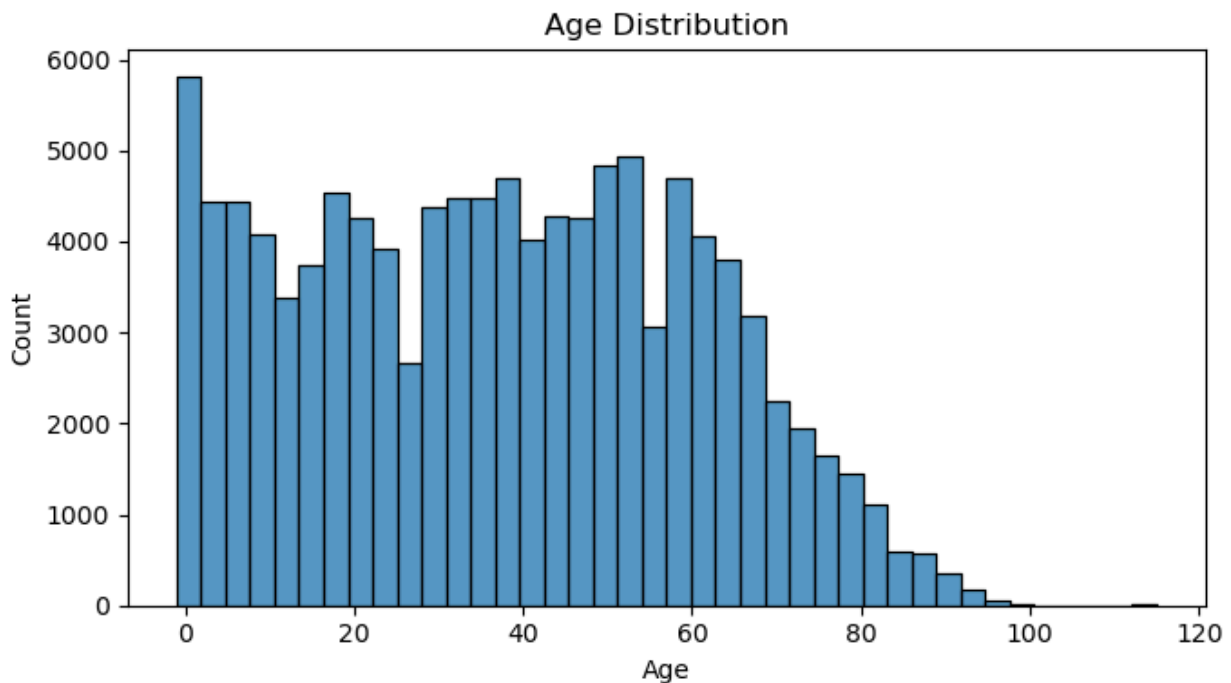


Figure 2. Age distribution of patients in the dataset.

The age distribution shows that the majority of patients fall within the adult age range, with fewer patients at extremely young or extremely old ages. The distribution appears moderately skewed, reflecting variations in healthcare utilization across different age groups.

Understanding the age distribution is important because age is expected to influence healthcare behavior. Younger and older patients may have different healthcare needs, priorities, and appointment attendance patterns.

The distribution of the outcome variable, No-show, was also examined. The outcome variable represents whether a patient attended their appointment or missed it. A count plot was generated to visualize the frequency of each outcome category.

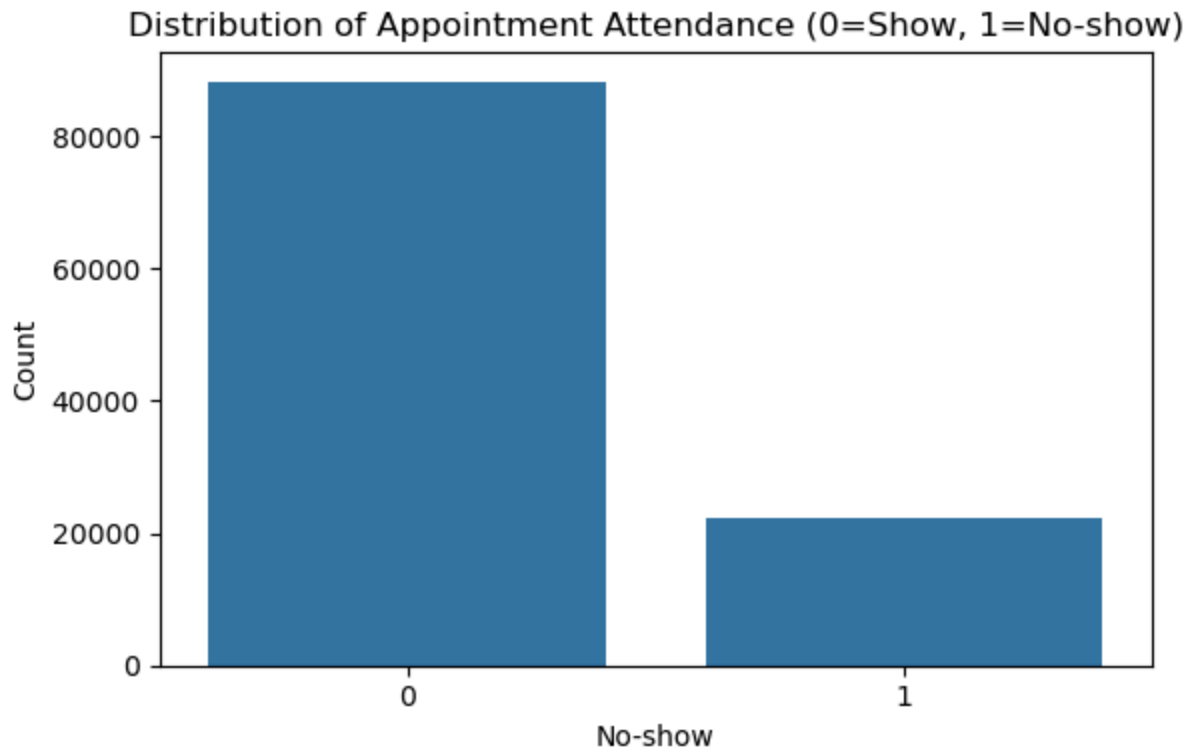


Figure 3. Distribution of appointment attendance outcomes.

Analysis of the outcome variable showed that approximately 79.8% of patients attended their scheduled appointments, while approximately 20.2% of patients missed their appointments. This indicates that the dataset is moderately imbalanced, with a higher number of attended appointments compared to missed appointments.

Class imbalance is an important consideration in predictive modeling because it can influence model performance. Models may achieve high accuracy simply by predicting the majority class. Therefore, additional evaluation metrics such as ROC curve and confusion matrix analysis are necessary to properly assess model performance.

The relationship between Age and appointment attendance was also examined. Boxplots were used to compare the distribution of age between patients who attended appointments and those who missed appointments.

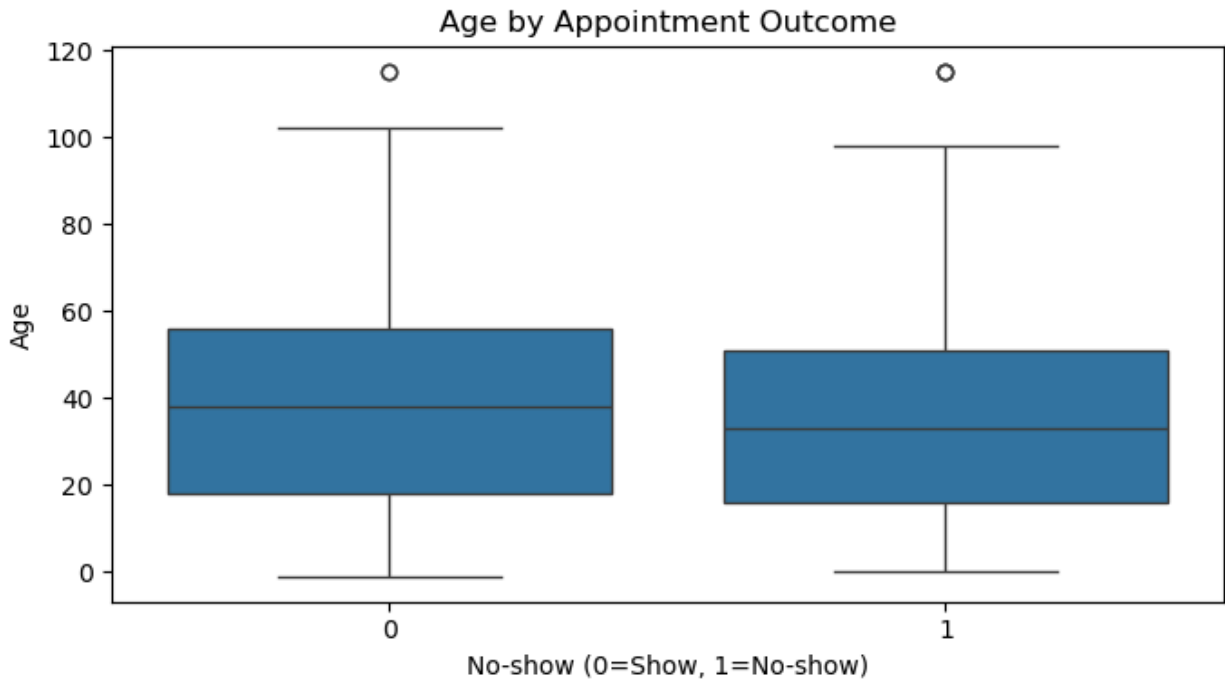


Figure 4. Age distribution by appointment outcome.

The boxplot shows that patients who missed appointments tend to have slightly lower median age compared to patients who attended appointments. This observation suggests that age may be associated with appointment attendance behavior.

The exploratory analysis provided important insights into the dataset. The dataset contains a large and diverse sample of patient appointments, with meaningful variation in predictor variables. The outcome variable shows moderate class imbalance, and predictor variables such as Age appear to be associated with appointment attendance behavior.

Exploratory data analysis also confirmed that the dataset is suitable for predictive modeling, with properly formatted variables and sufficient sample size to support reliable statistical inference and machine learning model training.

Overall, exploratory data analysis provided a strong foundation for subsequent statistical and machine learning modeling, enabling a deeper understanding of the dataset and informing the selection of appropriate modeling techniques.

Statistical Analysis and Logistic Regression

Statistical analysis was conducted to formally evaluate the relationship between predictor variables and the appointment attendance outcome. Statistical methods provide a rigorous framework for testing hypotheses, quantifying relationships between variables, and determining whether observed differences are statistically significant. In this study, statistical analysis focused on evaluating whether patient age is significantly associated with appointment no-show behavior and quantifying the strength of this relationship using logistic regression.

The first step in the statistical analysis involved hypothesis testing to determine whether there was a statistically significant difference in age between patients who attended appointments and those who missed appointments. Hypothesis testing is used to evaluate whether observed differences between groups are likely due to random variation or represent a meaningful relationship.

The null hypothesis (H_0) was defined as follows:

H_0 : There is no statistically significant difference in age between patients who attended appointments and patients who missed appointments.

The alternative hypothesis (H_1) was defined as follows:

H_1 : There is a statistically significant difference in age between patients who attended appointments and patients who missed appointments.

To test this hypothesis, an independent samples t-test was performed. The independent samples t-test compares the mean values of two independent groups to determine whether they are statistically different. In this case, the two groups consisted of patients who attended appointments and patients who missed appointments.

The results of the t-test showed a statistically significant difference in mean age between the two groups. The calculated t-statistic was approximately 20.09, and the corresponding p-value was extremely small ($p < 0.001$). Since the p-value is much smaller than the commonly used significance level of 0.05, the null hypothesis was rejected.

This result indicates that age is statistically significantly associated with appointment attendance behavior. Specifically, the mean age of patients who attended appointments was higher than the mean age of patients who missed appointments. This finding suggests that younger patients are more likely to miss appointments compared to older patients.

To further quantify the relationship between age and appointment attendance, logistic regression analysis was performed. Logistic regression is a statistical modeling technique used to estimate the probability of a binary outcome based on one or more predictor variables. In this study, logistic regression was used to model the probability that a patient would miss an appointment based on their age.

The logistic regression model is defined by the following equation:

$$P(Y = 1) = 1 / (1 + e^{-(\beta_0 + \beta_1 X)})$$

where:

$P(Y = 1)$ represents the probability of a patient missing the appointment,

β_0 represents the intercept term,

β_1 represents the coefficient associated with the age predictor,

X represents the age of the patient.

The logistic regression model was estimated using maximum likelihood estimation, which identifies the parameter values that maximize the probability of observing the given data.

Logit Regression Results						
Dep. Variable:	No-show	No. Observations:	110527			
Model:	Logit	Df Residuals:	110525			
Method:	MLE	Df Model:	1			
Date:	Sun, 22 Feb 2026	Pseudo R-squ.:	0.003636			
Time:	18:00:40	Log-Likelihood:	-55401.			
converged:	True	LL-Null:	-55603.			
Covariance Type:	nonrobust	LLR p-value:	6.166e-90			
	coef	std err	z	P> z	[0.025	0.975]
const	-1.1377	0.014	-82.956	0.000	-1.165	-1.111
Age	-0.0066	0.000	-20.015	0.000	-0.007	-0.006

Figure 5. Logistic regression results showing the relationship between age and appointment no-show probability.

The logistic regression results showed that the coefficient for age was negative and statistically significant ($p < 0.001$). This indicates that as age increases, the probability of missing an appointment decreases. In other words, older patients are less likely to miss appointments compared to younger patients.

The statistical significance of the age coefficient confirms that age is an important predictor of appointment attendance. The negative coefficient indicates an inverse relationship between age and no-show probability. This finding is consistent with the exploratory data analysis and hypothesis testing results.

Logistic regression provides both predictive capability and interpretability. Unlike machine learning models, logistic regression allows researchers to directly interpret the relationship between predictor variables and the outcome. This interpretability is particularly important in healthcare settings, where understanding the factors influencing patient behavior is essential for designing effective interventions.

The logistic regression model achieved a predictive accuracy of approximately 79.93%, indicating strong predictive performance using only a small number of predictor variables. This result demonstrates that traditional statistical models remain highly effective for healthcare predictive modeling.

Overall, the statistical analysis and logistic regression results confirm that age is a statistically significant and meaningful predictor of appointment attendance behavior. These findings provide a strong statistical foundation for the machine learning models evaluated in the next section.

Machine Learning Models and Model Evaluation

In addition to statistical modeling, machine learning models were implemented to evaluate predictive performance and compare results with the logistic regression model. Machine learning algorithms are capable of capturing complex relationships between predictor variables and outcomes and are widely used in healthcare predictive analytics.

In this study, two machine learning algorithms were implemented: Random Forest and Extreme Gradient Boosting (XGBoost). These models were selected due to their strong predictive performance and ability to identify important predictor variables.

Random Forest is an ensemble learning method that constructs multiple decision trees and combines their predictions to improve classification accuracy and reduce overfitting. Each decision tree is trained on a random subset of the data, and the final prediction is determined by aggregating the predictions of all trees.

Extreme Gradient Boosting (XGBoost) is an advanced machine learning algorithm based on gradient boosting. XGBoost builds models sequentially, where each new model improves upon the errors of the previous model. XGBoost is widely used due to its efficiency, scalability, and strong predictive performance.

The machine learning models were trained using the same predictor variables used in the logistic regression model, including Age, Gender, Scholarship status, Hypertension status, Diabetes status, and SMS reminder status. The dataset was divided into training and testing sets to evaluate model performance on unseen data.

Model performance was evaluated using confusion matrix analysis, Receiver Operating Characteristic (ROC) curve analysis, and feature importance analysis.

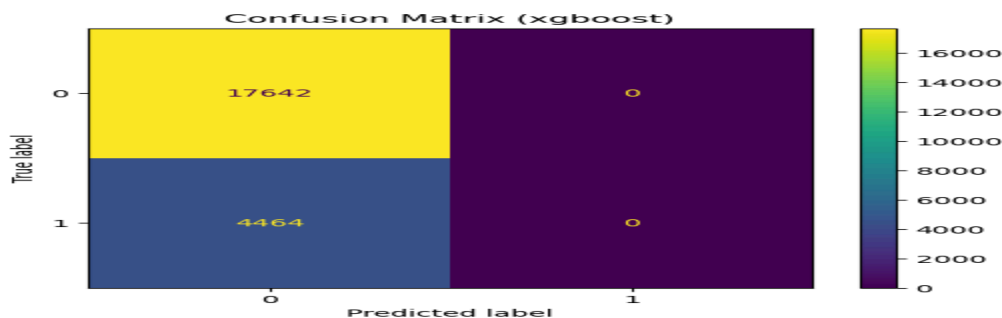


Figure 6. Confusion matrix for the XGBoost model.

Write this paragraph below caption:

The confusion matrix shows the classification performance of the XGBoost model. The model correctly classified a large number of patients who attended appointments. However, the model did not correctly classify patients who missed appointments. This result indicates that the model is biased toward predicting the majority class due to class imbalance in the dataset. This highlights the importance of using evaluation metrics beyond simple accuracy.

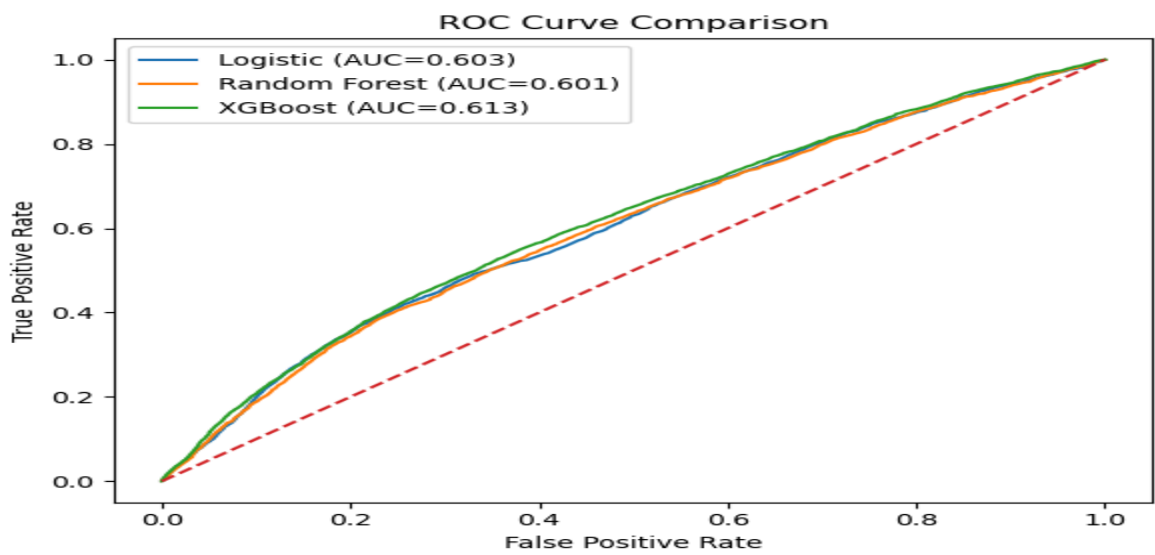


Figure 7. ROC curve comparison of Logistic Regression, Random Forest, and XGBoost models.

The ROC curve evaluates the ability of each model to distinguish between patients who attended appointments and those who missed appointments. The Area Under the Curve (AUC) values for Logistic Regression, Random Forest, and XGBoost were approximately 0.603, 0.601, and 0.613, respectively. These values indicate moderate predictive performance. The similarity in performance across models suggests that machine learning models do not significantly outperform the statistical model for this dataset.

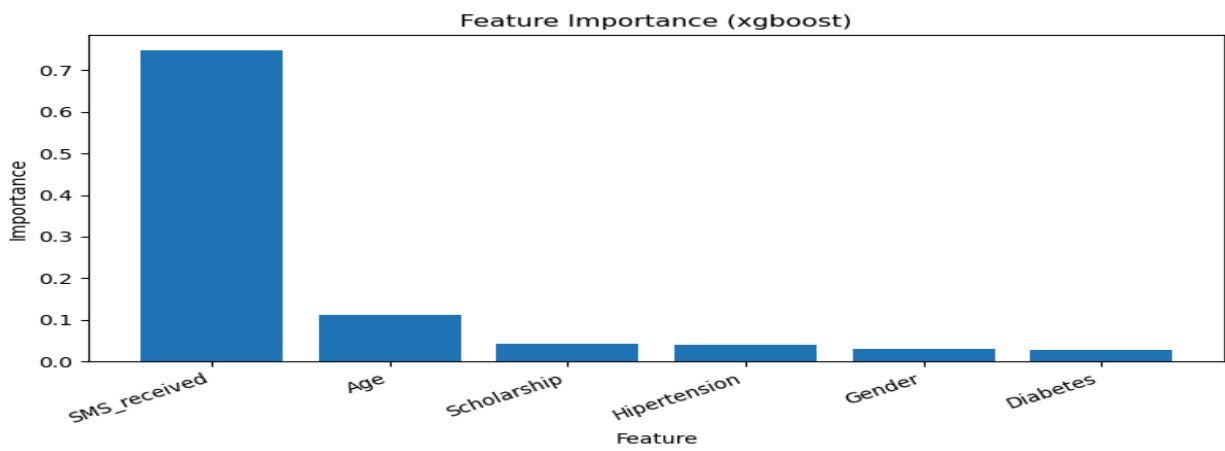


Figure 8. Feature importance analysis using the XGBoost model.

Feature importance analysis shows that SMS_received is the most important predictor of appointment attendance. This indicates that patients who receive SMS reminders are significantly more likely to attend their appointments. Age is the second most important predictor, suggesting that demographic factors also influence appointment attendance. Other variables such as scholarship status, hypertension, gender, and diabetes have relatively smaller predictive importance.

Overall, machine learning models achieved similar predictive performance compared to the logistic regression model. While machine learning models provide advantages in identifying feature importance and capturing complex relationships, the results demonstrate that traditional statistical models remain highly effective for predicting healthcare appointment attendance. These findings also highlight the importance of communication-based interventions, such as SMS reminders, in improving patient attendance.

Discussion

This study evaluated the effectiveness of statistical and machine learning methods for predicting patient appointment no-show outcomes using healthcare operational data. The results demonstrated that both statistical modeling and machine learning approaches provide meaningful insights into patient attendance behavior, with comparable predictive performance across models.

The statistical analysis confirmed that age is a statistically significant predictor of appointment attendance. Hypothesis testing showed a highly significant difference in mean age between patients who attended appointments and those who missed appointments. Logistic regression analysis further confirmed that age has a negative and statistically significant association with appointment no-show probability. This finding indicates that younger patients are more likely to miss appointments compared to older patients. This relationship may reflect differences in healthcare utilization patterns, lifestyle factors, and prioritization of healthcare appointments across age groups.

The machine learning models implemented in this study, including Random Forest and XGBoost, achieved predictive performance similar to the logistic regression model. The ROC curve analysis showed Area Under the Curve (AUC) values ranging from approximately 0.60 to 0.61 across models. These results indicate moderate predictive performance and suggest that the available predictor variables capture some, but not all, of the factors influencing appointment attendance behavior.

The similarity in predictive performance between statistical and machine learning models highlights an important finding. While machine learning models are capable of capturing complex relationships, they do not always outperform simpler statistical models, particularly when the dataset contains a limited number of predictor variables. In this study, logistic regression achieved performance comparable to more complex machine learning models, demonstrating the continued relevance and effectiveness of traditional statistical modeling approaches in healthcare analytics.

Feature importance analysis provided valuable insights into the factors influencing appointment attendance. The analysis identified SMS reminder status as the most important predictor of appointment attendance. This finding has important practical implications for healthcare providers. SMS reminders represent a simple and cost-effective intervention that can significantly improve appointment attendance rates. Healthcare providers can use predictive models to identify high-risk patients and prioritize reminder interventions for those patients.

The confusion matrix results showed that the model correctly identified most patients who attended appointments but struggled to correctly identify patients who missed appointments. This limitation is likely due to class imbalance in the dataset, where the number of attended appointments is substantially higher than the number of missed appointments. Class imbalance

is a common challenge in healthcare predictive modeling and can reduce the ability of models to accurately identify minority class outcomes.

Another important limitation of this study is the limited number of predictor variables available in the dataset. Appointment attendance behavior is influenced by a wide range of factors, including socioeconomic status, transportation availability, employment status, patient engagement, and health literacy. These factors were not available in the dataset and may improve predictive performance if included in future models.

Despite these limitations, this study provides valuable insights into healthcare predictive modeling. The results demonstrate that both statistical and machine learning models can be used to predict appointment attendance and identify important predictor variables. The study also highlights the importance of operational variables, particularly SMS reminders, in improving healthcare appointment attendance.

This research contributes to the growing field of healthcare analytics by providing a direct comparison between statistical and machine learning approaches using real-world healthcare data. The findings demonstrate that statistical models remain highly effective and interpretable tools for healthcare prediction, while machine learning models provide additional insights through feature importance analysis.

Future research can improve predictive performance by incorporating additional predictor variables, using larger datasets, and exploring more advanced modeling techniques. Healthcare providers can use predictive modeling to identify patients at risk of missing appointments and implement targeted interventions to improve healthcare system efficiency.

Conclusion

This study conducted a comprehensive analysis of statistical and machine learning methods for predicting patient appointment no-show outcomes using healthcare operational data. The objective of this research was to evaluate the predictive performance of logistic regression, Random Forest, and Extreme Gradient Boosting (XGBoost) models and identify key factors influencing appointment attendance behavior.

The statistical analysis demonstrated that age is a statistically significant predictor of appointment attendance. Hypothesis testing confirmed a significant difference in age between patients who attended appointments and those who missed appointments. Logistic regression analysis further showed that age has a negative relationship with appointment no-show probability, indicating that younger patients are more likely to miss appointments compared to older patients.

The machine learning models implemented in this study achieved predictive performance comparable to the statistical model. The ROC curve analysis showed Area Under the Curve (AUC) values ranging from approximately 0.60 to 0.61, indicating moderate predictive ability. These results demonstrate that machine learning models do not significantly outperform logistic regression for this dataset, suggesting that statistical models remain highly effective for healthcare predictive modeling.

Feature importance analysis identified SMS reminder status as the most important predictor of appointment attendance. This finding highlights the importance of communication-based interventions in improving healthcare operational efficiency. Healthcare providers can use predictive models to identify patients at higher risk of missing appointments and implement targeted reminder systems to improve attendance rates.

The results of this study demonstrate that predictive modeling can provide valuable insights into patient behavior and healthcare system efficiency. Both statistical and machine learning approaches offer useful tools for healthcare analytics, with statistical models providing strong interpretability and machine learning models offering additional insights into predictor importance.

This study has several limitations. The dataset contains a limited number of predictor variables, and additional factors such as socioeconomic status, transportation access, and patient engagement were not included. Future research can improve predictive performance by incorporating additional variables and exploring more advanced modeling techniques.

Despite these limitations, this research demonstrates the practical value of predictive modeling in healthcare analytics. The findings highlight the continued relevance of statistical modeling and the complementary role of machine learning methods in predicting patient appointment attendance.

In conclusion, this study provides a comprehensive comparison of statistical and machine learning approaches for healthcare appointment prediction. The results demonstrate that logistic regression, Random Forest, and XGBoost models provide meaningful predictive insights and can be used to support healthcare operational decision-making. Predictive modeling represents a valuable tool for improving healthcare efficiency and patient outcomes.

References

Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32.
<https://doi.org/10.1023/A:1010933404324>

Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794.
<https://doi.org/10.1145/2939672.2939785>

Hosmer, D. W., Lemeshow, S., & Sturdivant, R. X. (2013). *Applied Logistic Regression* (3rd ed.). Wiley.

James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An Introduction to Statistical Learning* (2nd ed.). Springer.

Kuhn, M., & Johnson, K. (2013). *Applied Predictive Modeling*. Springer.

Goldstein, B. A., Navar, A. M., Pencina, M. J., & Ioannidis, J. P. A. (2017). Opportunities and challenges in developing risk prediction models with electronic health records data. *American Journal of Epidemiology*, 187(8), 1787–1794.

Topol, E. J. (2019). High-performance medicine: The convergence of human and artificial intelligence. *Nature Medicine*, 25(1), 44–56.